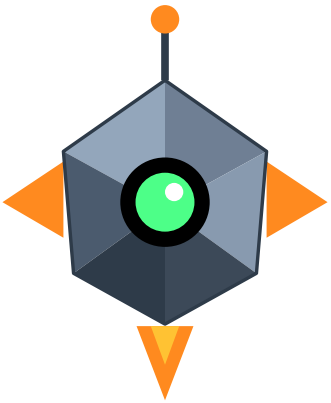
BOOK 8

The Boundary of a Step

Counting steps, and what they cost





MULTIX // HELLO
I am Multix. I ride along in every chapter. Watch my readout: it shows what is owed, what is paid, and what is only a guess.

CHAPTER 1

One Step, Many Steps



MULTIX // CIRCLE CHECK
A circle holds a piece that belongs together, and circles can sit inside circles. To **pop** a ready circle, replace it with its value and leave everything outside it alone. In this book, a circle is *ready* when everything needed for its calculation is known and its operation is on the menu. To **redraw**, move circles around or split a number, as long as the value stays the same. The full rules are in the introduction.

STORY

Ibo and Sade were having a race. The question was **6 × 7**, and the winner would be whoever finished in the fewest steps.
Ibo said, “One step. It’s 42.”
Sade said, “Five steps. I did 7 + 7 + 7 + 7 + 7 + 7.”
Their cousin Tomas laughed. “I counted up from zero, one number at a time, until I hit 42. That’s forty-two steps.”
All three got the same answer. All three told the truth about how many steps they took. So who won?

STEP 1 Three ways to count

The trouble is that nobody had agreed what counts as *one step*.

Ibo’s way: Ibo knows his times tables, so “6 × 7 = 42” is one fact, one step.

Sade’s way: Sade added 7 to itself. Six sevens need five additions: 7 + 7 = 14, then 21, 28, 35, 42. Each addition is one step, so five steps.

Tomas’s way: Tomas counted by ones. Counting from 0 to 42 is forty-two steps.

Same problem, same answer, and three different step counts: 1, 5 and 42. Nobody cheated. They just drew the boundary of a step in three different places.

STEP 2 Big steps hide small ones

Ibo’s one step is really a bundle. Behind it sits Sade’s five additions. Behind each addition, if you went even smaller, sits a bit of counting.

Trust check. A comparison of step counts is trustworthy only if everyone used **the same menu**.

A calculator does this too. When you press the × button, the little chip inside carries out many tiny steps in a blink. You count it as one step, because you cannot see the tiny ones and you do not need to.

So "how many steps?" really means *"how many steps, if these are the kinds of step we allow?"*

IN CIRCLES

A **step** is one pop. The menu says which circles count as *ready*: a circle is ready when everything needed for its calculation is known and its operation is on the menu.

With times tables on the menu, Ibo's circle is ready at once:

TRACE

```
( 6 × 7 )  →  42        1 pop
```

With only addition on the menu, (6 × 7) is not ready, so Sade redraws it first:

TRACE

```
( 6 × 7 )
redraw  →  ( ( ( ( ( 7 + 7 ) + 7 ) + 7 ) + 7 ) + 7 )
pop ( 7 + 7 )   →  14
pop ( 14 + 7 )  →  21
pop ( 21 + 7 )  →  28
pop ( 28 + 7 )  →  35
pop ( 35 + 7 )  →  42   5 pops
```

Same circle, same value, a different number of pops. The menu decides how many.

TWO COUNTS

Look at Sade's trace again. It has 5 pops and 1 redraw. There are two honest ways to count it.

- **Calculation steps** count only the pops: Sade has 5, and Ibo has 1.
- **Recorded moves** count every line of the trace, including redraws and refusals: Sade has 6, and Ibo has 1.

Neither count is wrong. They answer different questions. The first asks how much arithmetic was done, and the second asks how much the notebook had to record. A race needs one of them chosen and stated, and the choice can change who looks fastest.

STEP 3 A menu of steps

Imagine a game where only three kinds of step are allowed:

double, **halve** and **add**. That is the whole menu. You may not just multiply.

Can you find **5 × 9** with that menu?

- Double 9 to get 18. (1 step)
- Double 18 to get 36. (2 steps)
- Now 36 is four 9s. You need five, so add one more 9: 36 + 9 = 45. (3 steps)

So 5 × 9 = 45 in three steps.

Now try **13 × 7**. Double 7 three times: 14, 28, 56. That is three steps, and 56 is eight 7s. You need thirteen 7s, and 13 is 8 + 4 + 1. So add 56 + 28 + 7:

56 + 28 = 84, then 84 + 7 = 91.

That is two more steps, so five steps in all. And 13 × 7 = 91.

With a different menu, the same problems would take a different number of steps. If multiplying is allowed, 13 × 7 is a single step. If only counting by ones is allowed, it takes ninety-one.

STEP 4 A fair race

Mina and Dev raced to find **99 × 17**.

Mina did: 100 × 17 = 1,700, then 1,700 − 17 = 1,683. She counted **2 steps**.

Dev did: 90 × 17 = 1,530, then 9 × 17 = 153, then 1,530 + 153 = 1,683. He counted **3 steps**.

Mina said she won. Dev was not so sure. “Is 100 × 17 really one step?” he asked. “I’d have to think about that one.”

Here is the honest answer: *it depends on the menu.* If every multiplication counts as one step, and so does every addition and subtraction, then Mina wins, 2 steps against 3. If the menu is different, the count may change.

So before any race, any contest or any claim that one solution is shorter, the first question is: **what counts as a step?** Write the menu down. Then count.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What stayed the same for Ibo, Sade and Tomas?** (The answer, 42.)
- **What changed?** (How many steps they counted. 1, 5 and 42.)
- **What was missing from the race?** (An agreement about what counts as a step.)

WHAT MAKES THIS COMPARISON TRUSTWORTHY?

A comparison of step counts is trustworthy only if everyone used **the same menu**. Counting Ibo's one big step against Tomas's forty-two tiny ones is like comparing a mile with a kilometer by looking only at the numbers 1 and 1.6. Different units, different counts.
The answer to "which is shorter?" can be reliable and fair, but only after the menu is stated.

PRACTICE

1. Count the steps for **8 × 6** in three ways: as a times-table fact, as repeated addition of 6s, and by counting up from 0 by ones.
2. Using only **double**, **halve** and **add**, find **5 × 11**. How many steps?
3. Using the same menu, find **9 × 7**. (Hint: 9 is 8 + 1.)
4. Mina says: "I took 2 steps and you took 3, so I win." What should Dev ask her before agreeing?
5. How many steps does **12 × 12** take if you add twelve 12s one at a time? Why might nobody want to use that method?
6. With only addition on the menu, how many pops does (4 × 9) take? Write the redrawn circle, and say how many recorded moves the trace has.

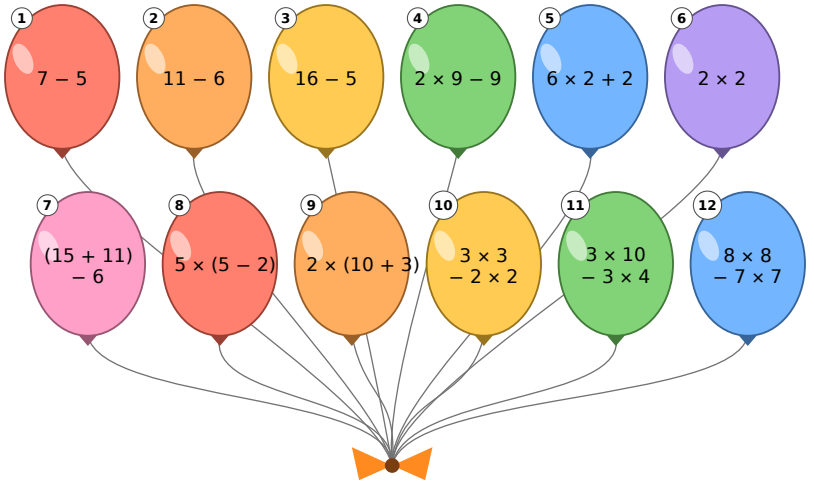
BRIDGE *Bridge to the rest of the series:* In The Choice Before the Calculation, choosing 100 as a reference cut the number of steps, but only under a particular menu. In Two Boxes, One Answer, the detective sees only the answer, never the number of steps inside. In Reversing the Calculation, the diary is written at a chosen level: "multiplied 6 by 7" is one line, while "added 7 five times" is five lines.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

											!
1	2	3	4	5	6	7	8	9	10	11	12

CHAPTER 2

The Step That Grows

STORY

Sade and Tomas were at it again. This time the problem was $\mathbf{3^4}$.
The little raised 4 tells you how many copies of 3 to multiply together. So 3^4 means $3 \times 3 \times 3 \times 3$. These are called *powers*, and 3^4 is read "3 to the power of 4."
Both cousins had the same plain calculator, with a × button and nothing else. They disagreed about what the button could be used on.
Tomas's rule: each step multiplies the number on the screen by 3.
Sade's rule: each step multiplies two numbers you already have.

STEP 1 Counting the steps

Tomas followed his rule:

3 × 3 = 9. Then 9 × 3 = 27. Then 27 × 3 = **81**.

That is 3 multiplications, so **3 steps**.

Sade started the same way: 3 × 3 = 9. Then she noticed she could use the 9 twice:

9 × 9 = **81**.

That is **2 steps**.

Multiplying a number by itself is called *squaring*. 9 squared is 81.

Same answer, fewer steps. Notice where the saving came from. It did not come from a special button. It came from reusing a result she had already worked out, the 9. Tomas could do exactly the same thing on his plain calculator once he spotted it. Sade's rule just allows it.

IN CIRCLES

Here are Tomas's and Sade's notebooks.

Trust check. Three things have to be true.

TRACE

```
( ( ( 3 × 3 ) × 3 ) × 3 )          Tomas
pop ( 3 × 3 )  →  9
pop ( 9 × 3 )  →  27
pop ( 27 × 3 )  →  81              3 pops
```

TRACE

```
( ( 3 × 3 ) × ( 3 × 3 ) )          Sade
pop ( 3 × 3 )  →  9     (use this 9 in both places)
pop ( 9 × 9 )  →  81               2 pops
```

Sade's notebook rule is this: once a circle has popped to a number, you may use that number again as often as you like. That is where her saving comes from. Tomas can reuse a value too, once he notices it.

STEP 2 A step that grows itself

Now the numbers get larger. Try the powers of 2, using a menu that has only two steps: **double** and **square**.

Starting at 2, how fast can we grow?

Square 2 and you get 4. Square 4 and you get 16. Square 16 and you get 256. Square 256 and you get **65,536**.

That is only **4 steps** to reach 65,536. For comparison, starting from the same 2, doubling over and over takes 15 steps, and counting up by ones takes 65,534 steps. Squaring grows so fast because each squaring uses the whole number as both ingredients.

Now aim at a specific number: $\mathbf{2^{10}}$, which is 1,024.

- Doubling only: start at 2 and double nine times (4, 8, 16, 32, 64, 128, 256, 512, 1,024). That is **9 steps**.
- With squaring allowed:
  - Square 2 to get 4. (1 step)
  - Square 4 to get 16. (2 steps)
  - Double 16 to get 32. (3 steps)
  - Square 32 to get **1,024**. (4 steps)

So 4 steps instead of 9.

STEP 3 Why squaring works

Look at the last move, squaring 32. Why does that give 2^{10}?

32 is 2^5, which means five 2s multiplied together. Squaring 32 means multiplying 32 by 32, which puts two piles of five 2s together. That makes ten 2s multiplied together. And ten 2s is 2^{10}.

So squaring *doubles the number of copies*. Doubling the number itself only adds *one* more copy. That is why squaring is the step that grows.

STEP 4 The menu decides

How many steps does 2^{10} take? It depends on the menu again, just as in chapter 1.

- Menu with only **double**: 9 steps.
- Menu with **double and square**: 4 steps.
- Menu with a **power button** on the calculator: 1 step.

All three are honest counts. They answer different questions.

For a power that is not just 2s, use the same idea. Try $\mathbf{3^5}$, with a menu of "multiply by 3" and "square":

- Square 3 to get 9. (1 step)
- Square 9 to get 81. (2 steps)
- Multiply 81 by 3 to get **243**. (3 steps)

Using only "multiply by 3," it would take 4 steps (9, 27, 81, 243).

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What stayed the same whichever menu we used?** (The answer: 81, or 1,024, or 243.)
- **What changed?** (The number of steps.)

WHAT MAKES THIS COUNT TRUSTWORTHY?

Three things have to be true.
First, the rewriting must be exact. $2^5 \times 2^5$ really is 2^{10}, because five 2s and five 2s together make ten 2s. That is a reason, not a guess.
Second, the saving must have a source we can name. Here it is reusing a result we already worked out, such as the 9 or the 32. Squaring is just the name for multiplying a number by itself.
Third, the step counts are only comparable under the same menu, as in chapter 1. Saying "4 steps" without saying "with double and square" is not a complete answer.

PRACTICE

1. Find $\mathbf{3^4}$ using only the rule "multiply the number on the screen by 3." Then find it again by multiplying a result by itself. How many steps each time?
2. Find $\mathbf{5^4}$ (which means $5 \times 5 \times 5 \times 5$) by squaring. How many steps?
3. Find $\mathbf{2^8}$ using the menu of **double** and **square**. How many steps? How many steps would it take with

doubling only?

4. Find $\mathbf{3^5}$ with a menu of “multiply by 3” and “square.” Check by multiplying 3 five times.

5. *Brave reader:* Can you reach 2^{10} in only **3 steps** with the menu of double and square? Try listing every number you can reach after each step.

6. Write 5^4 with circles so that one pop can be reused. How many pops does it take?

BRIDGE *Bridge to the rest of the series:* In The Work That Vanishes, squares were areas, and 50×50 gave the big square. Here the same idea shows up as a step that grows. In The Choice Before the Calculation, splitting 36 as 4×9 to work out 25×36 was a choice made before any arithmetic, and spotting that a result can be reused is a choice of the same kind.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 37 × 43 )
2 redraw  →  ( ( 40 − 3 ) × ( 40 + 3 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 3 × 3 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 3 × 3 )  →  12
6 pop ( 1,600 − 12 )  →  1,588
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 19 × 21 )
2 redraw  →  ( ( 20 − 1 ) × ( 20 + 1 ) )
3 redraw  →  ( ( 20 × 20 ) − ( 1 × 1 ) )
4 pop ( 20 × 20 )  →  400
5 pop ( 1 × 1 )  →  1
6 pop ( 400 − 1 )  →  399
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 64 × 76 )
2 redraw  →  ( ( 70 − 6 ) × ( 70 + 6 ) )
3 redraw  →  ( ( 70 × 70 ) − ( 6 × ( 6 + 1 ) ) )
4 pop ( 70 × 70 )  →  4,900
5 pop ( 6 × ( 6 + 1 ) )  →  42
6 pop ( 4,900 − 42 )  →  4,858
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 3

Three Diaries of the Same Calculation

STORY

Sade, Ibo and Tomas each calculated 12 × 13. Each kept a diary. They all got the right answer, but their diaries looked very different.

STEP 1 Three diaries

Ibo's diary had one line:

TRACE

```
( 12 × 13 )  →  156
```

Sade's diary had three lines:

TRACE

```
( 12 × 13 )
redraw  →  ( ( 12 × 10 ) + ( 12 × 3 ) )
pop both  →  ( 120 + 36 )
pop  →  156
```

Tomas's diary wrote every single pop on its own line:

TRACE

```
( 12 × 13 )
redraw  →  ( ( 12 × 10 ) + ( 12 × 3 ) )
pop ( 12 × 10 )  →  120
( 120 + ( 12 × 3 ) )
pop ( 12 × 3 )  →  36
( 120 + 36 )
pop  →  156
```

These three diaries describe the same calculation. They just zoom in to different levels. Ibo's zooms out so far that the whole calculation is one step. Tomas's zooms in so far that every pop is a step.

STEP 2 What does each diary let you see?

On Tuesday, a new student, Pip, calculated 12 × 13 and got 146. Pip's answer was wrong. The teacher wanted to find the mistake.

Trust check. A diary is trustworthy if each line can be checked by itself: the circles on the left and on the right have the same value.

If Pip had kept a diary like Ibo's:

TRACE

```
( 12 × 13 )  →  146
```

The mistake is somewhere, but the diary doesn't say where. All the teacher can do is repeat the whole calculation.

If Pip had kept a diary like Sade's:

TRACE

```
( 12 × 13 )
redraw  →  ( ( 12 × 10 ) + ( 12 × 3 ) )
pop both  →  ( 120 + 26 )
pop  →  146
```

Now the teacher can see it. 120 is right, but (12 × 3) is 36, not 26. The mistake is on the third line, in the "pop both".

If Pip had kept a diary like Tomas's, the teacher would see the exact line: *pop (12 × 3) → 26*.

A finer diary pins down a mistake more precisely.

STEP 3 But finer is not always better

Tomas's diary is the longest. For a hard calculation with hundreds of steps, a diary that shows every pop would fill a whole notebook, and nobody would read it. A reader could easily lose the main idea in the details.

So there is a trade. A finer diary tells you more about *where*, and costs more time to write and read. A coarser diary is quick and clear about the big idea, and hides the details.

The right level depends on the job.

- A shopper adding up a bill needs only Ibo's level.
- A teacher looking for a mistake wants Sade's or Tomas's.
- A person teaching the trick, to show why 12 × 13 splits into 12 × 10 and 12 × 3, wants Sade's.

STEP 4 A step is a choice of zoom

In the last two chapters, we said a step is one pop. That is Tomas's level. But the word "step" depends on the diary.

- At Ibo's level, 12 × 13 is **1 step**.
- At Sade's level, it is **3 steps**.
- At Tomas's level, it is **4 steps**: one redraw and three pops.

For this reason, whenever someone says “that took 3 steps,” a good question is **“3 steps at what level?”**. The answer is only meaningful when you know the zoom.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What do all three diaries keep?** (The question and the answer, 156.)
- **What does a finer diary keep that a coarser one drops?** (The place where a mistake could hide.)

WHAT MAKES THIS TRUSTWORTHY?

A diary is trustworthy if each line can be checked by itself: the circles on the left and on the right have the same value. The coarse diary is no less trustworthy in what it says, but it gives you less to check.
When you have two diaries of the same level that disagree, there is a mistake in at least one. You can compare line by line to find where they first differ.

PRACTICE

1. Write Sade’s level of diary for 15×12, splitting it as $(15 \times 10) + (15 \times 2)$.
2. Pip wrote this diary: $((15 \times 10) + (15 \times 2)) \to (150 + 20) \to 170$. Which line has the mistake? What should it be?
3. How many steps is 15×12 at Ibo’s level? At Sade’s level (your diary from problem 1)?
4. Name one job where a coarse diary is enough, and one where you would want a fine diary.
5. Two friends both say “it took 5 steps.” One used a diary that writes every pop, and the other used one that groups pops. Could they still have done the same calculation? Explain.

BRIDGE *Bridge to the rest of the series:* In chapter 1 of this book, a step was one pop on a menu. Here the zoom level is a second choice, and it decides what the word “step” means. In Reversing the Calculation, a diary is what keeps the road back, and a finer diary keeps more of it.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **2,491**. Leave the others blank. What appears?

2460 + 24	2466 + 21	2480 + 9	2522 − 31	2507 − 16	2460 + 25	2469 + 28	51 × 51 − 11 × 11
2464 + 18	2469 + 23	2481 + 10	50 × 50 − 3 × 3	2530 − 39	2470 + 21	2484 + 13	119 × 21
46 × 54	2459 + 32	2471 + 20	2519 − 28	2470 + 21	2499 − 8	2455 + 36	10 × 248
2504 − 13	2529 − 38	2477 + 14	2503 − 12	2514 − 23	2497 − 6	2511 − 20	2511 − 20
2483 + 8	2498 − 7	2513 − 22	2472 + 19	2494 − 3	2458 + 33	2499 − 8	2520 − 29
2458 + 33	2466 + 25	2512 − 21	2478 + 23	2477 + 9	2479 + 12	2489 + 2	2456 + 35
2466 + 25	2469 + 22	2506 − 15	2484 + 11	2493 + 8	2529 − 38	2504 − 13	2495 − 4
2510 − 19	2465 + 26	2483 + 8	2484 + 6	29 × 86	2469 + 22	2505 − 14	2495 − 4



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 4

Possible and Permitted

STORY

Sade ran the class shop. A customer bought 3 notebooks at 49 cents each and asked for the exact bill.
Sade's eyes went straight to the shortcut. "49 is nearly 50, and 3 × 50 is easy."
On a calculator she could press that button. It was a **possible** move. But was it **permitted**?

STEP 1 A move can be possible and still not allowed

The customer wanted the exact bill. Rounding 49 up to 50 changes the value, so it is not a fair move for an exact problem. Sade wrote a new kind of line in her notebook, a **refusal**. It records a move she could make, and why she did not.

TRACE

```
( 3 × 49 )
refuse ( 3 × 50 )  →  not allowed: the bill must be exact
```

Then she used a move that *was* allowed. She split 49 as 50 − 1, and corrected for it:

TRACE

```
( 3 × 49 )
redraw  →  ( ( 3 × 50 ) − ( 3 × 1 ) )
pop ( 3 × 50 )  →  150
pop ( 3 × 1 )  →  3
( 150 − 3 )
pop  →  147
```

Why is this redraw fine when rounding was not? Because the redraw kept the value. It used 3 × 50, and then took back the 3 × 1 that the 50 had added. Rounding alone would have lost that correction.

The exact bill is **147 cents**.

STEP 2 The same move, a different job

Suppose the customer had said, "Just give me a rough idea." Then (3 × 50) would have been permitted, because a rough idea only needs to be close. Sade could write:

Trust check. The work is trustworthy when every pop was possible and every move was permitted, and when the notebook says why.

TRACE

```
( 3 × 49 )
estimate ( 3 × 50 )  →  150    (rough answer, error 3)
```

The move did not change. The job changed, and the job decides what is permitted.

STEP 3 When a calculator says yes and the problem says no

Twelve cookies are shared among 5 children, and cookies cannot be broken.

A calculator will happily say 12 ÷ 5 = 2.4. That is a possible move with a possible answer. But it is not permitted by the problem, because nobody can have 2.4 whole cookies.

TRACE

```
( 12 ÷ 5 )
refuse 2.4  →  not allowed: cookies stay whole
each child gets 2, with 2 left over
```

STEP 4 Four kinds of moves

Possible and permitted are two separate questions. Sade made a table:

Move	Possible now?	Permitted by the problem?
(3 × 49) in an exact bill	yes	yes
Rounding 49 to 50 in an exact bill	yes	**no**
(x + 12) when the jar x is still sealed	**no, not yet**	yes
8 ÷ 0	no	no

The third row is the one to notice. A move can be allowed by the rules and still be impossible today, because something is missing. In What Must Remain Unresolved, that is what being stuck means: nothing is forbidden, it just cannot be done yet.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the refusal line keep?** (The reason, so a reader can see the rule that shaped the work.)
- **What did Sade keep by choosing the redraw?** (The exact value of the bill.)

WHAT MAKES THIS TRUSTWORTHY?

The work is trustworthy when every pop was possible and every move was permitted, and when the notebook says why. If someone asks, “Why didn’t you just round?”, the refusal line answers. Without it, the reader cannot tell a careful choice from a missed shortcut.

PRACTICE

1. Sade sells 4 pens at 98 cents each and the bill must be exact. Write the refusal line for (4 × 100), then find the bill with a redraw.
2. A shopper only wants a rough idea of the cost of the same 4 pens. Is (4 × 100) permitted now? How far off is it?
3. Seventeen marbles are shared among 4 friends, and marbles stay whole. What does a calculator give for 17 ÷ 4? Write a refusal line, and say what each friend gets and what is left.
4. Fill in “possible now?” and “permitted?” for each: (a) (9 + 5), (b) (n + 5) with n unknown, (c) rounding 18 to 20 in an exact sum.
5. Write a refusal line for rounding 19 to 20 in the exact bill (3 × 19), then find the bill another way.

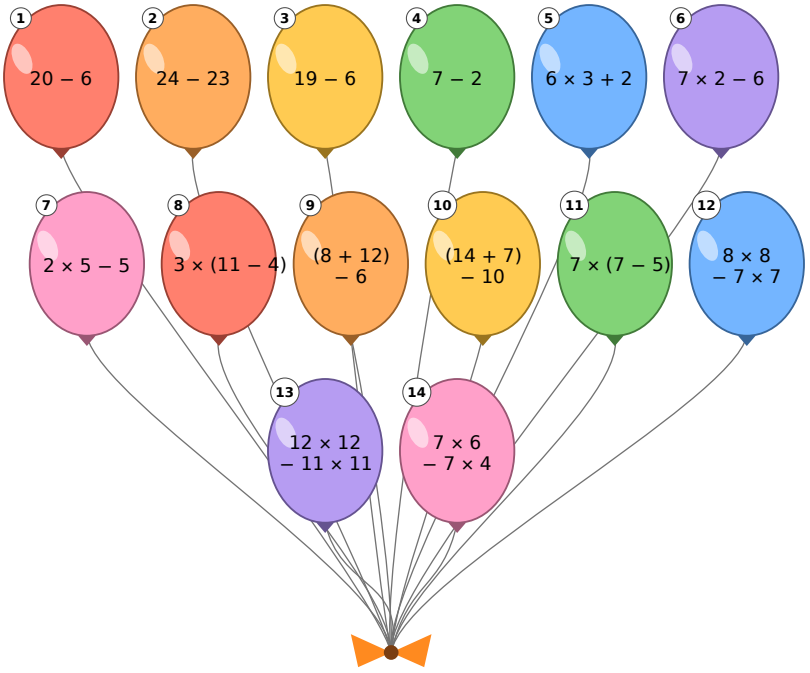
BRIDGE *Bridge to the rest of the series:* In Solving by Changing the Question, the whole-cookie rule is a refusal: a change of question that ignores it gives an answer that does not fit. In A Useful Wrong Answer, an estimate is permitted exactly when the error is named.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

													!
1	2	3	4	5	6	7	8	9	10	11	12	13	14

CHAPTER 5

The Cost of the Idea

STORY

Ibo wanted to add every number from 1 to 100. He counted steps in two ways.

STEP 1 One number at a time

The slow way adds each number to a running total.

TRACE

```
( 1 + 2 )  →  3
( 3 + 3 )  →  6
( 6 + 4 )  →  10
...
```

Each line is one pop of an addition, which is on the menu. How many are there? There are 100 numbers. The first line uses two of them, and every line after that adds one more. So there are **99 additions**.

STEP 2 The pairing way

In The Order of Discovery, Kofi paired the ends. The first and last make 101, and there are 50 such pairs.

TRACE

```
( 1 + 100 )  →  101
( 50 × 101 )  →  5,050
```

That is **2 pops**: one addition and one multiplication, both on the menu.

99 steps against 2 steps. It looks like a huge difference.

Trust check. A step count is trustworthy when it says what the menu is and says what ideas were used.

STEP 3 But the idea was not free

Ibo then asked a fair question: *what did it cost to think of pairing?*

To write down those 2 pops, someone had to notice that the ends pair up, that every pair makes the same sum, and that there are exactly 50 pairs. None of that was counted as a step. It was an **idea**, and the idea took thinking.

So “2 steps” really means “2 steps on the menu, after one idea that is not on the menu.” Even the recorded moves miss

it. A redraw into pairs is one line in the trace, but the thinking that led to it is in neither count.

This is a lesson about counting. There are two honest ways to say how hard something is.

- **Steps on the menu**: how many pops once you know what to do.
- **The idea**: what you had to see before you could write the pops down.

For adding 1 to 100, the idea is valuable because it works for every list like it. Once Kofi has it, the next list costs two steps and no new idea.

STEP 4 The idea pays for itself

Try the same idea on 1 to 1,000.

- One at a time: **999 additions**.
- By pairs: (1 + 1,000) → 1,001, then (500 × 1,001) → 500,500. That is **2 pops**.

The slow way grew with the list. The pairing way did not. A good idea costs thought once and saves steps every time after.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did both methods keep?** (The sum.)
- **What did the step count leave out?** (The thinking that came before the pops.)

WHAT MAKES THIS TRUSTWORTHY?

A step count is trustworthy when it says what the menu is and says what ideas were used. “2 steps” alone is incomplete. “2 steps on this menu, after the idea of pairing” is honest. The sum itself is checked by the other method: 5,050 for 1 to 100 either way.

PRACTICE

1. How many additions does it take to add 1 to 10 one at a time? How many pops with pairing?
2. How many additions does it take to add 1 to 20 one at a time?

3. Use pairing to add 1 to 1,000. What is the sum?

4. Ibo says: “Pairing is 2 steps, so it is easy.” What would you say to him?

5. Name another idea that cost thought once and saved steps later.

BRIDGE *Bridge to the rest of the series:* In chapter 3, the same calculation was written at different levels of detail. Here the question is what a count leaves out entirely. In The Work That Vanishes, the missing corner is an idea that costs thought once and saves work afterward.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 38 × 42 )
2 redraw  →  ( ( 40 − 2 ) × ( 40 + 2 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 2 × ( 2 + 1 ) ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 2 × ( 2 + 1 ) )  →  6
6 pop ( 1,600 − 6 )  →  1,594
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 24 × 26 )
2 redraw  →  ( ( 25 − 1 ) × ( 25 + 1 ) )
3 redraw  →  ( ( 25 × 25 ) − ( 1 × 1 ) )
4 pop ( 25 × 25 )  →  625
5 pop ( 1 × 1 )  →  1
6 pop ( 625 − 1 )  →  624
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 46 × 54 )
2 redraw  →  ( ( 50 − 4 ) × ( 50 + 4 ) )
3 redraw  →  ( ( 50 × 50 ) − ( 4 × 4 ) )
4 pop ( 50 × 50 )  →  2,500
5 pop ( 4 × 4 )  →  20
6 pop ( 2,500 − 20 )  →  2,480
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 6

What a Check Costs

STORY

Sade had worked out 23 × 47 and got 1,081. Before she wrote it on the bill, she wanted to check it. But checks cost steps too.

STEP 1 Three checks and their cost

Check 1, the last digit. The last digits are 3 and 7, and 3 × 7 = 21, so the answer should end in 1.

TRACE

```
( 3 × 7 )  →  21
```

1 pop. 1,081 ends in 1. ✓

Check 2, the size. The answer must lie between two easy products.

TRACE

```
( 20 × 40 )  →  800
( 25 × 50 )  →  1,250
```

2 pops. 1,081 is between 800 and 1,250. ✓

Check 3, a second route. Split 47 as 40 + 7.

TRACE

```
( ( 23 × 40 ) + ( 23 × 7 ) )
pop  →  ( 920 + 161 )
pop  →  1,081
```

3 pops. It agrees. ✓

STEP 2 Cost against what each check can catch

Check	Cost	What it can catch
Last digit	1 pop	slips in the ones digit
Size	2 pops	answers far too big or too small
Second route	3 pops	almost any slip

Trust check. A check is trustworthy when you know what it can catch and what it cannot.

The cheapest check catches the fewest mistakes. The most expensive one catches the most, but it costs about as much as doing the whole problem again.

A sensible plan, which Sade followed, is to **start cheap**. Last digit first. If it fails, the answer is wrong and she saves the other steps. If it passes, she tries the size check. And she keeps the second route for answers that matter.

STEP 3 What counts as a step in a check

Remember from chapter 5 that a step count leaves out the ideas. The last-digit check needs an idea too: that the ones digit of a product only depends on the ones digits. Once you know it, it costs 1 pop.

So a check has two prices:

- **The pops** it uses each time.
- **The idea** you need to know first, which you pay once.

STEP 4 When a check is not worth it

If the bill is for a few cents and an error would not matter, even the last-digit check may not be worth it. If the bill is for a thousand dollars, the second route is a bargain.

The right amount of checking depends on how much a mistake would cost. That is a decision about the job, not a rule of arithmetic.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did all three checks keep?** (The question: is 1,081 right for 23 × 47?)
- **What differed?** (How much each could catch, and how many pops each used.)

WHAT MAKES THIS TRUSTWORTHY?

A check is trustworthy when you know what it can catch and what it cannot. A passed last-digit check says nothing about the tens. As in A Useful Wrong Answer, a passed check never proves an answer right, and a failed check proves it wrong.

PRACTICE

1. For 38 × 52 = 1,976, do the last-digit check. How many pops?
2. Do a size check for 38 × 52 using 38 × 50 and 40 × 52. Does 1,976 pass?
3. Do a second route for 38 × 52 by splitting 52 as 50 + 2. How many pops?
4. Put the three checks in order from cheapest to most expensive.
5. Sade's answer fails the last-digit check. Should she do the size check next? Why or why not?

BRIDGE *Bridge to the rest of the series:* In A Useful Wrong Answer, chapter 3 introduced these checks by what they catch. Here they are compared by what they cost. In The Choice Before the Calculation, picking a method is a choice that balances ease against risk, and so is picking a check.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **1,599**. Leave the others blank. What appears?

1588 + 3	114 × 14	1595 + 12	1596 + 3	1629 − 30	1582 + 25	1599 + 11	1581 + 7
1578 + 23	1589 + 12	1568 + 31	1624 − 25	1621 − 22	1605 − 6	1586 + 22	1579 + 25
1579 + 17	1570 + 29	1627 − 28	1565 + 34	1587 + 12	1582 + 17	1566 + 33	1566 + 27
13 × 123	1583 + 16	1613 − 14	1566 + 33	1627 − 28	1565 + 34	1603 − 4	1573 + 26
1582 + 17	1572 + 27	1572 + 27	1632 − 33	1627 − 28	1574 + 25	1590 + 9	40 × 40 − 1 × 1
1623 − 24	1578 + 21	1568 + 31	1573 + 14	1578 + 10	1597 + 2	1610 − 11	1575 + 24
1577 + 22	1606 − 7	1585 + 14	1583 + 20	1575 + 14	1591 + 8	1632 − 33	1634 − 35
1582 + 17	40 × 40 − 1 × 1	1605 − 6	1580 + 25	1571 + 27	1570 + 29	1612 − 13	1575 + 24



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 7

Your Menu Grows

STORY

Ibo, Sade and Tomas each did the sum 12 × 12 and wrote how many pops it took.

STEP 1 Three menus

Tomas knows his times tables up to 12 by heart. For him, 12 × 12 is on the menu.

TRACE

```
( 12 × 12 )  →  144
```

1 pop.

Sade knows her tables up to 10. She needs to split.

TRACE

```
( 12 × 12 )
redraw  →  ( ( 12 × 10 ) + ( 12 × 2 ) )
pop  →  ( 120 + 24 )
pop  →  144
```

3 pops (and one redraw).

Ibo is newer to multiplying. He only trusts adding for now, so he adds twelve twelves.

TRACE

```
12 + 12 → 24 → 36 → 48 → 60 → 72 → 84 → 96 → 108 → 120 → 132 → 144
```

11 pops.

Three people, three counts: 1, 3 and 11. The sum is the same. The menus are different.

STEP 2 A menu is what you can do in one step

In The Boundary of a Step, we said a step is one pop of something on the menu. Now we see that the menu belongs to a person. It holds whatever that person can do in one go.

Trust check. A step count is trustworthy when it says what the menu is.

- For a beginner, the menu has addition.
- Later it adds the times tables.

- Later it adds tricks, such as the missing corner.

Each time something new joins the menu, the same problem takes fewer steps.

STEP 3 How a menu grows

Sade learned a new fact: $12 \times 12 = 144$. She did it by practice, over a few weeks. Now it is on her menu.

Did the calculation get simpler? Not for the problem. 12×12 is still the same product. What changed is Sade. What used to take three pops now takes one.

This is an important thing to say honestly. When someone says "that was easy," it may mean the problem is easy, or it may mean they have done it many times. A beginner and an expert can both be right about their own step counts.

STEP 4 Saying whose menu

So when you count steps, say whose menu you are using.

- "11 steps, for someone who only adds."
- "3 steps, with the times tables to 10."
- "1 step, if 12×12 is known."

This is the idea from chapter 5 again: a count with no menu named is incomplete.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What stayed the same for all three?** (The product, 144.)
- **What was different?** (How many steps, because the menus were different.)

WHAT MAKES THIS TRUSTWORTHY?

A step count is trustworthy when it says what the menu is. Without that, "it took 3 steps" does not say whether the problem is easy or the person is experienced.
The product is checked the same way for everyone: 12×12 is 144 because $144 \div 12 = 12$.

PRACTICE

1. Count the pops to find 7×8 for a person who only adds. For a person who knows the times tables.

2. Sade knows tables to 10. How many pops does she need for 13 × 6? (Split 13.)
3. Tomas knows tables to 12. How many pops does 12 × 11 take for him?
4. A beginner says “13 × 6 is hard” and an expert says it is easy. Can both be right?
5. Name something that was once hard for you and is now one step.

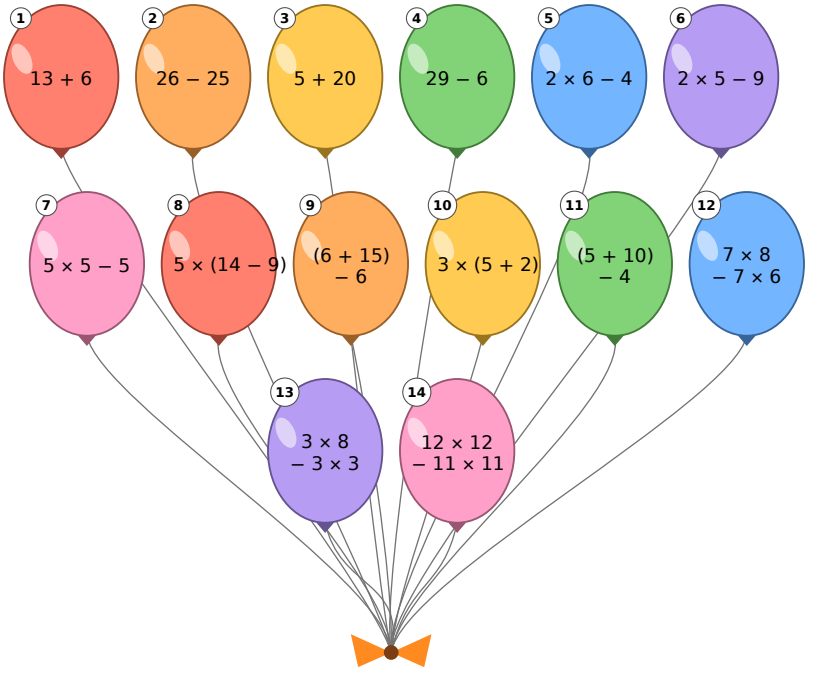
BRIDGE *Bridge to the rest of the series:* In chapter 5, the cost of an idea was left out of a step count. Here the cost of practice is left out too, and a growing menu is how it shows up. In The Order of Discovery, ideas that once took a messy notebook can become a single step.
Next in this book: Chapter 8: The Step You Never Take.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

													!
1	2	3	4	5	6	7	8	9	10	11	12	13	14

CHAPTER 8

The Step You Never Take

STORY

Kit had two helpers, and a card with a very strange instruction: **write down the first number and throw away the second.**
The card said:

TRACE

```
( keep-first  7  ( 987 × 654 ) )
```

A **keep-first** circle has two slots. It keeps what is in the first slot and throws away what is in the second. Here the first slot is 7, and the second holds a big multiplication.

STEP 1 The eager helper

The eager helper follows one rule: *finish everything inside before touching the outside.*

TRACE

```
pop ( 987 × 654 )  →  645,498
pop ( keep-first 7 645,498 )  →  7
```

Two pops, one of them a large multiplication, and the answer is 7. All the work on the 645,498 was thrown away at once.

STEP 2 The lazy helper

The lazy helper looks at the keep-first circle first. The rule of a keep-first circle is that the second slot is thrown away, so it never needs to be opened.

Trust check. Skipping work is trustworthy when you can give the reason.

TRACE

```
pop ( keep-first 7 ( 987 × 654 ) )  →  7      second slot never opened
```

One pop, the same answer, and no multiplication at all.

By the ready rule, the keep-first circle is ready, because the calculation it needs is only the first slot. The second slot is not needed.

Notice that the lazy helper did not skip a step carelessly. It looked at the circle, saw that the second slot could not change

the result, and left it alone. The trace says so: *second slot never opened*.

STEP 3 A slot that never finishes

Kit then tried a worse card.

TRACE

```
( keep-first  4  ( count every whole number ) )
```

The second slot is an endless job. Counting every whole number never finishes. The eager helper starts on it first and never gets to the outside. It counts forever.

The lazy helper pops the outside circle at once and answers **4**.

Both helpers were doing legal moves. The only difference was the order, and one order arrived while the other never did. The answer, 4, was there all along.

STEP 4 Lazy is not always better

Kit tried a card where the second slot *is* used.

TRACE

```
( 5 + ( 987 × 654 ) )
```

Here the multiplication is part of the sum, so it cannot be skipped. Both helpers must finish it. The lazy helper is only quicker when a piece will be thrown away.

Kit also noticed something about the case file. In 48 × 52, nothing is thrown away. The correction of 4 is needed. But if someone only asked, “is 48 × 52 more than 2,000?”, the lazy answer would not take the correction step at all. 2,500 minus any small corner is clearly still above 2,000, and that is the step you never take.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the lazy helper keep?** (The answer, 7 or 4, which was the same as the eager helper’s.)
- **What did it avoid?** (Work on a piece that would be thrown away.)

WHAT MAKES THIS TRUSTWORTHY?

Skipping work is trustworthy when you can give the reason. Here the reason is the rule of the circle: the second slot cannot change the result. If you cannot give a reason, do not skip. Kit marks a skip as **has a reason** only when it comes with the rule that justifies it.

PRACTICE

1. Find (keep-first 12 (345 × 678)) with the lazy helper. How many pops? How many for the eager helper?
2. Find ((keep-first 9 (77 × 88)) + 1) with the lazy helper. How many pops?
3. Can the lazy helper skip the multiplication in (3 + (12 × 12))? Why or why not?
4. Which helper answers (keep-first 8 (count every whole number))? What is the answer?
5. Someone says, “The lazy helper is less careful.” What do you say?

BRIDGE *Bridge to the rest of the series:* In Two Boxes, One Answer, a note on the fridge saves work by reusing a value, and here a thrown-away slot saves work by never computing it. In The Order of Discovery, two routes that both finish always agree, but one route may never finish.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 37 × 43 )
2 redraw  →  ( ( 40 − 3 ) × ( 40 + 3 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 3 × 3 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 3 × 3 )  →  9
6 pop ( 1,600 − 9 )  →  1,591
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 19 × 21 )
2 redraw  →  ( ( 20 − 1 ) × ( 20 + 1 ) )
3 redraw  →  ( ( 20 × 20 ) − ( 1 × ( 1 + 1 ) ) )
4 pop ( 20 × 20 )  →  400
5 pop ( 1 × ( 1 + 1 ) )  →  2
6 pop ( 400 − 2 )  →  398
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 64 × 76 )
2 redraw  →  ( ( 70 − 6 ) × ( 70 + 6 ) )
3 redraw  →  ( ( 70 × 70 ) − ( 6 × 6 ) )
4 pop ( 70 × 70 )  →  4,900
5 pop ( 6 × 6 )  →  42
6 pop ( 4,900 − 42 )  →  4,858
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

Answers (turn the book around)

Chapter 1

1. As a fact: **1** step. As repeated addition: eight 6s need seven additions, so **7** steps. By counting from 0 to 48: **48** steps.
2. Double 11 to get 22, double 22 to get 44, then add 11 to get 55 (44 is four 11s, and 5 = 4 + 1). That is **3 steps**. 5 × 11 = 55.
3. Double 7 three times: 14, 28, 56. That is three steps, and 56 is eight 7s. Add one more 7: 56 + 7 = 63. That is a fourth step, so **4 steps**. 9 × 7 = 63.
4. Dev should ask: "What counts as one step?" If a multiplication, an addition and a subtraction each count as one step, Mina's 2 beats Dev's 3. But they should agree on the menu before counting.
5. Twelve 12s need eleven additions, so **11 steps**. Nobody wants to use this method because the same answer is available in one step, if you know the fact.
6. Redraw it as (((9 + 9) + 9) + 9). The pops give 18, then 27, then 36, so it takes **3 pops**. With the redraw included, the trace has **4 recorded moves**.

Balloon Message.

1.	$7 - 5$	$= 2$	B
2.	$11 - 6$	$= 5$	E
3.	$16 - 5$	$= 11$	K
4.	$2 \times 9 - 9$	$= 9$	I
5.	$6 \times 2 + 2$	$= 14$	N
6.	2×2	$= 4$	D
7.	$(15 + 11) - 6$	$= 20$	T
8.	$5 \times (5 - 2)$	$= 15$	O
9.	$2 \times (10 + 3)$	$= 26$	Z
10.	$3 \times 3 - 2 \times 2$	$= 5$	E
11.	$3 \times 10 - 3 \times 4$	$= 18$	R
12.	$8 \times 8 - 7 \times 7$	$= 15$	O

Message: **BE KIND TO ZERO!**

Chapter 2

1. With "multiply by 3" only: **3 steps** (9, 27, 81). By multiplying a result by itself: **2 steps** (9, then 9 × 9 = 81).
2. 5 × 5 = 25, then 25 × 25 = **625**. That is **2 steps**.
3. Square 2 to get 4, square 4 to get 16, square 16 to get **256**. That is **3 steps**. With doubling only, it takes **7 steps** (4, 8, 16, 32, 64, 128, 256).
4. $3^2 = 9$, $9^2 = 81$, 81 × 3 = **243**. That is **3 steps**. Check: 3 × 3 × 3 × 3 × 3 = 243.
5. **No.** After 1 step, the only number you can reach is 4 (doubling and squaring 2 both give 4). After 2 steps you can reach 8 or 16. After 3 steps you can reach 16, 64, 32 or 256. Since 1,024 is not on that list, it cannot be reached in 3 steps. The 4-step route is the shortest with this menu.
6. ((5 × 5) × (5 × 5)). Pop (5 × 5) to get 25 and use it in both places, and then (25 × 25) pops to **625**. That is **2 pops**.

Spot the Slip.
Notebook A: slip on line **5**. 3 × 3 is 9. The answer should be 1,591.
Notebook B: **perfect**. 19 × 21 = 399.
Notebook C: slip on line **3**. The corner is 6 × 6, not 6 × (6 + 1). The answer should be 4,864.

Answers (turn the book around)

Chapter 3

1. (15 × 12), then redraw to ((15 × 10) + (15 × 2)), then pop both to (150 + 30), then pop to **180**.
2. The mistake is in "(15 × 2)", which is 30, not 20. The line should be (150 + 30), giving 180.
3. At Ibo's level, **1 step**. At Sade's level, **3 lines** after the question (redraw, pop both, pop).
4. Many answers work. A coarse diary is enough for a quick shopping bill. A fine diary is better for finding an error in a test.
5. Yes. "Steps" depends on how finely you write the diary, so the same calculation can be 5 steps at one level and 3 at another. Only the same level compares fairly.

Color by Answer (2,491). The shaded boxes make a house.



Chapter 4

1. `refuse ( 4 × 100 ) → not allowed: the bill must be exact.` Then ((4 × 100) − (4 × 2)) pops to (400 − 8), which is **392** cents.
2. Yes, because a rough idea only needs to be close. It is 400 against the exact 392, so it is off by **8**.
3. A calculator gives 4.25. `refuse 4.25 → not allowed: marbles stay whole.` Each friend gets **4**, with **1** left over.
4. (a) possible, permitted. (b) not possible yet, permitted. (c) possible, **not** permitted.
5. `refuse ( 3 × 20 ) → not allowed: the bill must be exact.` Then ((3 × 20) − (3 × 1)) pops to (60 − 3), which is **57** cents.

Balloon Message.

1.	$20 - 6$	$= 14$	N
2.	$24 - 23$	$= 1$	A
3.	$19 - 6$	$= 13$	M
4.	$7 - 2$	$= 5$	E
5.	$6 \times 3 + 2$	$= 20$	T
6.	$7 \times 2 - 6$	$= 8$	H
7.	$2 \times 5 - 5$	$= 5$	E
8.	$3 \times (11 - 4)$	$= 21$	U
9.	$(8 + 12) - 6$	$= 14$	N
10.	$(14 + 7) - 10$	$= 11$	K
11.	$7 \times (7 - 5)$	$= 14$	N
12.	$8 \times 8 - 7 \times 7$	$= 15$	O
13.	$12 \times 12 - 11 \times 11$	$= 23$	W
14.	$7 \times 6 - 7 \times 4$	$= 14$	N

Message: **NAME THE UNKNOWN!**

Answers (turn the book around)

Chapter 5

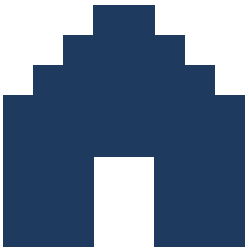
1. One at a time: **9 additions**. With pairing: (1 + 10) → 11 and (5 × 11) → 55, which is **2 pops**.
2. There are 20 numbers, so **19 additions**.
3. 500 pairs of 1,001, so **500,500**.
4. Two steps once you know the idea. The idea itself is part of the cost, and it should be counted or named.
5. For example, the missing corner: seeing it once makes products like 48 × 52 easy afterward. Other answers work too.

Spot the Slip.
Notebook A: slip on line **3**. The corner is 2 × 2, not 2 × (2 + 1). The answer should be 1,596.
Notebook B: **perfect**. 24 × 26 = 624.
Notebook C: slip on line **5**. 4 × 4 is 16. The answer should be 2,484.

Chapter 6

1. 8 × 2 = 16, so the answer should end in 6. 1,976 ends in 6. ✓ **1 pop**.
2. 38 × 50 = 1,900 is below the answer because 52 is more than 50, and 40 × 52 = 2,080 is above it because 38 is less than 40. 1,976 is between them. ✓ **2 pops**.
3. ((38 × 50) + (38 × 2)) → (1,900 + 76) → **1,976**. **3 pops**. ✓
4. Last digit (1 pop), size (2 pops), second route (3 pops).
5. No. A failed check already proves the answer is wrong. She should redo the problem.

Color by Answer (1,599). The shaded boxes make a house.



Answers (turn the book around)

Chapter 7

1. For adding only: seven eights take **6 additions** (or eight sevens, 7 additions). With the tables: **1 pop**.
2. ((10 × 6) + (3 × 6)) → (60 + 18) → 78. That is **3 pops** (and one redraw).
3. He knows it: (12 × 11) → 132, **1 pop**.
4. Yes. The problem is the same, but the menus differ. "Hard" depends on whose menu.
5. Many answers work, such as a times table fact, a spelling, or tying shoes.

Balloon Message.

1.	13 + 6	= 19	S
2.	26 − 25	= 1	A
3.	5 + 20	= 25	Y
4.	29 − 6	= 23	W
5.	2 × 6 − 4	= 8	H
6.	2 × 5 − 9	= 1	A
7.	5 × 5 − 5	= 20	T
8.	5 × (14 − 9)	= 25	Y
9.	(6 + 15) − 6	= 15	O
10.	3 × (5 + 2)	= 21	U
11.	(5 + 10) − 4	= 11	K
12.	7 × 8 − 7 × 6	= 14	N
13.	3 × 8 − 3 × 3	= 15	O
14.	12 × 12 − 11 × 11	= 23	W

Message: **SAY WHAT YOU KNOW!**

Chapter 8

1. Lazy: **1** pop, answer 12. Eager: **2** pops (the multiplication, then the keep-first), answer 12.
2. Pop the keep-first to get 9, then pop (9 + 1). That is **2** pops, and the answer is **10**.
3. **No.** The 12 × 12 is added in, so it affects the result. It must be finished.
4. The **lazy** helper. The eager helper never finishes the count. The answer is **8**.
5. The lazy helper is careful about one thing: it only skips a piece when the rule says the piece cannot matter. Skipping with a reason is not careless.

Spot the Slip.
Notebook A: **perfect**. 37 × 43 = 1,591.
Notebook B: slip on line **3**. The corner is 1 × 1, not 1 × (1 + 1). The answer should be 399.
Notebook C: slip on line **5**. 6 × 6 is 36. The answer should be 4,864.